# COVER

## FLEET OVERVIEW & DRAWING INDEX

ARCTICCHAIN RURAL NODE FAMILY — COMPLETE PRINT SET

**AF-000**

REV A · 2026-11-12

PROTOTYPE

## THE FLEET — ONE NETWORK, MANY SHAPES

| Node | Codename | Role | Envelope | Power | Status |
|---|---|---|---|---|---|
| [illegible] | [illegible] | mailbox+repeater+bulletin+SOS | | | |
| [illegible] | [illegible] | public E-Ink + mesh node | | | |
| [illegible] | [illegible] | perchable env sensor | | | |
| [illegible] | [illegible] | — dock+relay+backhaul | | | |
| [illegible] | [illegible] | mesh-native, sat SOS | | | |
| [illegible] | [illegible] | — seasonal sensing | | | |
| [illegible] | [illegible] | payload — mesh IS ground segment | | | |

## SHARED DESIGN DNA (EVERY NODE)

- Climate: -45..+70 °C · IP65+ · UV-stable · LiFePO4-preferred · December-sized solar
- Power: solar-first · 30–90 day autonomy · works with zero grid
- Field: bear-country mounts · two-person install · hand-tool repair
- Sovereignty: community-owned mesh · data stays home · OCAP · EN/FR · privacy by design
- Appropriate tech: right-sized cost · repairable > sealed · open docs
- Offline dignity: works when everything else has failed — that IS the product

## DRAWING INDEX

| | |
|---|---|
| AF-000 | this sheet — fleet overview & index |
| AF-L1-101…104 | L1 Brick — GA, mechanical, electrical, software/assembly |
| AF-S-101…104 | S Mite — GA, mechanical, electrical, firmware/assembly |
| AF-E-101…103 | E Totem — GA, mechanical, electrical/content/software |
| AF-A-101…103 | A Anchor — GA, mechanical/power, electrical/backhaul |
| AF-P-101…102 | P ArcticPhone — concept + spec sketch |
| AF-D-101 | D Seed — R&D brief (degradable motes) |
| AF-H-101 | H Meridian — satellite mission concept |

## HOW TO USE THIS SET

- L1 Brick sheets are build-ready — real BOM, wiring, assembly, acceptance test.
- S Mite sheets are prototype-ready — orderable parts, CAD scripted (cad/swarm_node.py).
- E Totem / A Anchor sheets are spec-level — real parts selected, needs CAD pass.
- P Phone / D Seed / H Meridian are concept/R&D — honest status marked on every sheet.
- Every sheet: SURAFEL NEGUSSIE drew it · EMEBET INC. owns it · PROPRIETARY & CONFIDENTIAL.

| ARCTICCHAIN · RURAL NODE FLEET | | |
|---|---|---|
| PROPRIETARY & CONFIDENTIAL — EMEBET INC. | SOLE PROPRIETARY OWNER | DWG AF-000 |
| DRAWN: Surafel Negussie — Planner / Maker | Emebet Inc. · emebetos.ca registered Canadian company · Edmonton, AB | REV A |
| DESIGNED FOR RURAL & REMOTE CANADA — -45..+70 °C · IP65 · SOLAR-FIRST · COMMUNITY-OWNED | | SCALE N/S · SHEET 1/1 |

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